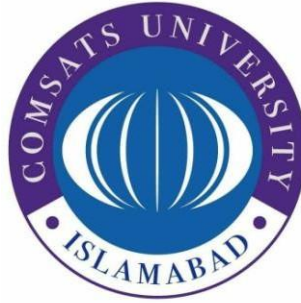




Department Of Computer Science

<i>Course</i>	<i>I-S LAB</i>
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<i>Assignment No.</i>	<i>01</i>

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CODE:

```
1  # Encryption Function
2  def caesar_encrypt(text, shift): 1 usage
3      result = ""
4
5      for char in text:
6          # Check if character is uppercase letter
7          if char.isupper():
8              result += chr((ord(char) - ord('A') + shift) % 26 + ord('A'))
9
10         # Check if character is lowercase letter
11         elif char.islower():
12             result += chr((ord(char) - ord('a') + shift) % 26 + ord('a'))
13
14         # Keep spaces and special characters same
15         else:
16             result += char
17
18     return result
19
```

```
20
21 # Decryption Function
22 def caesar_decrypt(ciphertext, shift): 1 usage
23     result = ""
24
25     for char in ciphertext:
26         if char.isupper():
27             result += chr((ord(char) - ord('A') - shift) % 26 + ord('A'))
28
29         elif char.islower():
30             result += chr((ord(char) - ord('a') - shift) % 26 + ord('a'))
31
32         else:
33             result += char
34
35     return result
36
```

```
37
38 # Main Program
39 text = input("Enter message: ")
40 shift = int(input("Enter shift value: "))
41
42 encrypted = caesar_encrypt(text, shift)
43 print("Encrypted Message:", encrypted)
44
45 decrypted = caesar_decrypt(encrypted, shift)
46 print("Decrypted Message:", decrypted)
```

EXPLANATION OF THE CODE:

The function `caesar_encrypt` accepts a message and a shift value. It initializes an empty string called `result` which stores encrypted characters. The loop iterates through each character in the message. Uppercase characters are detected using `isupper()` and converted using ASCII operations. The `ord()` function converts characters into numeric ASCII values. Subtracting `ord('A')` normalizes indexing from zero. The shift value moves characters forward. The modulo operator ensures wrap around when the alphabet exceeds Z. The `chr()` function converts ASCII numbers back into characters. Lowercase characters follow the same process using `ord('a')`. Spaces and special symbols are preserved using the `else` block. The `caesar_decrypt` function performs reverse transformation by subtracting the shift. The main program collects user input, encrypts the message, prints the encrypted text, then decrypts it to restore the original message.

SECURITY ANALYSIS:

The Caesar Cipher provides very limited security because the key space contains only 26 possible shifts. Attackers can easily recover plaintext using brute force testing. Additionally, frequency analysis reveals patterns because letter distributions remain unchanged after encryption. Modern encryption algorithms such as AES and RSA rely on complex mathematics and large keys to resist attacks. Therefore, Caesar Cipher is primarily suitable for learning cryptography fundamentals rather than protecting sensitive information.